Climate Informatics 2026 (2026), xx: 1–2
doi:10.1017/xxxx

CAMBRIDGE UNIVERSITY PRESS

CLIMATE INFORMATICS 2026
EXTENDED ABSTRACT

Unifying context-specific causal discovery in the PCMCI-framework with applications to river catchment data

Wiebke Günther[1], Oana-Iuliana Popescu[2] and Jakob Runge[2]

[1]Technische Universität Berlin, Institute of Computer Engineering and Microelectronics, Berlin, Germany
[2]Department of Computer Science, University of Potsdam, Potsdam, Germany

Received xx xxx xxxx

Keywords: causal discovery, context-dependent data, temporal and spatial regimes

1. Introduction and objectives

In many real-world systems, causal relationships vary across temporal or spatial contexts as underlying mechanisms change. In hydrology, precipitation–discharge links depend on soil moisture, while temperature effects vary with elevation and snowmelt. Accordingly, context-specific causal discovery has gained attention. Existing approaches differ in goals and assumptions: some aim to identify a *union graph* capturing causal relations valid across all contexts [2] and references therein, while others target regime- or context-specific graphs given known or inferred context assignments [3]. Within the time-series domain, several PCMCI-based methods have been proposed, including J-PCMCI$^+$ [2] and adaptive-context variants [3], addressing latent context confounding or endogenous context dynamics. However, selecting an appropriate method is not always straightforward.

Here, we unify existing PCMCI-based approaches within a modular framework for time-series causal discovery under context dependence. We clarify which algorithmic components are appropriate under different assumptions on context observability, exogeneity, and structure, and show that J-PCMCI$^+$ and adaptive-context- (AC-) PCMCI$^+$ arise as special cases. As an application, we study causal drivers of river discharge across heterogeneous European catchments [1].

2. Context-specific causal discovery in the PCMCI framework

We assume a context-dependent structural causal model where causal mechanisms may vary across time or space. Observed system variables depend on past system variables, observed context variables, and potentially unobserved exogenous context variables with structured spatio-temporal dependencies. Context variables may be endogenous or exogenous drivers of regime changes.

We build on the PCMCI$^+$ framework, which performs causal discovery via lagged parent discovery, Momentary Conditional Independence (MCI) testing, and an orientation phase. J-PCMCI$^+$ extends PCMCI$^+$ by pooling data across observed context variables, and introducing time- and space-dummy variables to account for latent context confounding, yielding a union graph under exogenous contexts. Alternatively, PCMCI$^+$ can be combined with an adaptive conditional-independence testing strategy [3] to directly recover context-specific causal graphs. MCI tests are performed on pooled data unless context variables appear in the conditioning set, in which case testing is restricted to context-specific subsets. This supports endogenous context variables while avoiding unnecessary tests.

© The Authors(s) 2020. Published by Cambridge University Press. This is an Open Access article, distributed under the terms of the Creative Commons Attribution licence (http://creativecommons.org/licenses/by/4.0/), which permits unrestricted re-use, distribution, and reproduction in any medium, provided the original work is properly cited.

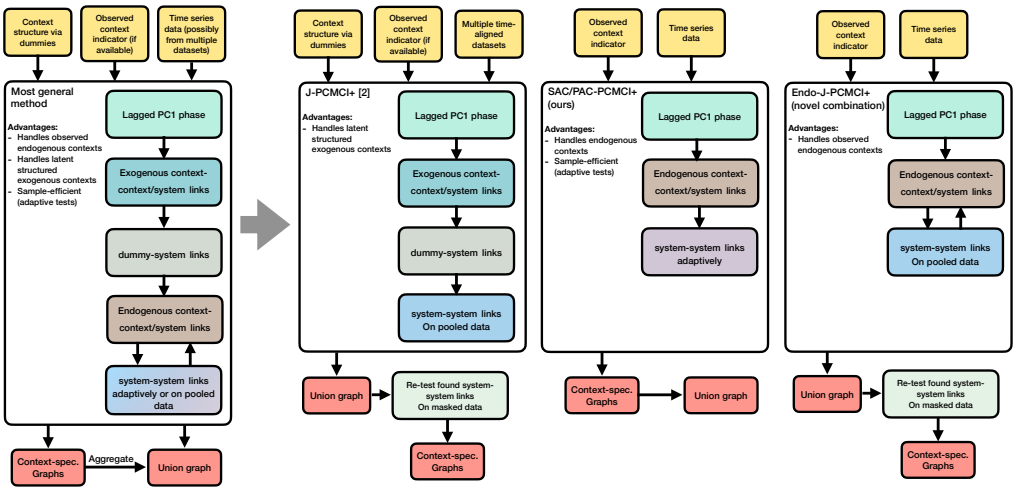

Figure 1. *General framework (left) encompassing [2] and [3] as special cases, allowing tailored methods for different settings. Example: Endo-J-PCMCI discovers a union graph with endogenous context variables (right).*

J-PCMCI$^+$ and AC-PCMCI$^+$ address complementary aspects of context-specific causal discovery, relying on different assumptions about context observability and exogeneity. The differences are subtle, but can significantly impact the outcomes, as violating certain assumptions of context properties can lead to false positives. We therefore propose a general, modular PCMCI-based framework that clarifies which components to activate under which assumptions, focusing on the MCI phase. Depending on assumptions, links involving exogenous contexts, latent structured contexts, and endogenous contexts are tested using pooled or context-restricted data, see Fig. 1.

3. Preliminary results and expected outcomes

We analyze data from 358 European river catchments [1]. Since available context variables are exogenous, we apply the J-PCMCI$^+$ variant of our framework and compute both context-specific and union graphs. The context variable is defined as a thresholded combination of catchment altitude and area.

Preliminary results show a stronger temperature-discharge link in high-altitude contexts, consistent with snowmelt processes. Future work will study exogenous latent and endogenous observed contexts. Limitations include reliance on context quality and remaining identifiability questions. Similar context dependence in socioeconomic and technical systems suggests broader applicability.

Acknowledgments. W.G. and J.R. received funding from the European Research Council (ERC) Starting Grant CausalEarth under the EU Horizon 2020 research and innovation program (Grant Agreement No. 948112). O.P. and J.R. received EU Horizon 2020 research and innovation program funding under grant agreement No 101003469 (XAIDA). This work used resources of the Deutsches Klimarechenzentrum (DKRZ) granted by its Scientific Steering Committee (WLA) under project ID 1083.

References

[1] W. Günther, P. Miersch, U. Ninad, and J. Runge, "Clustering of causal graphs to explore drivers of river discharge," *Environmental Data Science*, vol. 2, pp. 1–9, 2023.

[2] W. Günther, U. Ninad, and J. Runge, "Causal discovery for time series from multiple datasets with latent contexts," in *Proceedings of the Conference on Uncertainty in Artificial Intelligence*, pp. 766–776, 2023.

[3] W. Günther, O.-I. Popescu, M. Rabel, U. Ninad, A. Gerhardus, and J. Runge, "Causal discovery with endogenous context variables," *Advances in Neural Information Processing Systems*, vol. 37, pp. 36243–36284, 2024.